EXPERIMENT 5

AIM: **Write a Program to implement on a data set of characters the three CRC polynomials – CRC-12, CRC -16 and CRC-CCITT.**

SOLUTION:

CRC (Cyclic Redundancy Check) is an **error detection technique** used in the Data Link Layer to detect errors during data transmission. It uses a **generator polynomial (divisor)** to perform **Modulo-2 (XOR)** division on the data bits. The remainder obtained from the division is called the **CRC remainder**, which is appended to the original data before transmission. At the receiver, the received codeword is divided by the same generator polynomial. If the remainder is **all zeros**, the data is accepted; otherwise, an **error is detected**.

CRC	Polynomial	CRC size
CRC-12	$G(x) = x^{12} + x^{11} + x^3 + x^2 + x + 1$	12 bits
CRC-16	$G(x) = x^{16} + x^{15} + x^2 + 1$	16 bits
CRC-CCITT	$G(x) = x^{16} + x^{12} + x^5 + 1$	16 bits

Source Code:

```c
#include <stdio.h>
#include <string.h>
#include <conio.h>

void main()
{
    char data[100], divisor[20], temp[100], remainder[20];
    char received[100];
    int dataLen, divLen;
    int i, j;
    int error = 0;

    clrscr();

    printf("Enter the data bits: ");
    scanf("%s", data);

    printf("Enter the divisor (generator): ");
    scanf("%s", divisor);

    dataLen = strlen(data);
    divLen = strlen(divisor);

    /* Copy data into temp */
    strcpy(temp, data);

    /* Append zeros */
    for(i = 0; i < divLen - 1; i++)
    {
        temp[dataLen + i] = '0';
    }

    temp[dataLen + divLen - 1] = '\0';
```

```
    /* Sender Side CRC Calculation */
    for(i = 0; i < dataLen; i++)
    {
        if(temp[i] == '1')
        {
            for(j = 0; j < divLen; j++)
            {
                temp[i + j] = ((temp[i + j] - '0') ^ (divisor[j] - '0')) +
'0';
            }
        }
    }

    /* Extract CRC remainder */
    for(i = 0; i < divLen - 1; i++)
    {
        remainder[i] = temp[dataLen + i];
    }

    remainder[divLen - 1] = '\0';

    printf("\nCRC Remainder      : %s", remainder);
    printf("\nTransmitted Codeword : %s%s", data, remainder);

    /* Receiver Side */

    printf("\n\nEnter the received codeword: ");
    scanf("%s", received);

    strcpy(temp, received);

    dataLen = strlen(received);

    /* CRC Checking */

    for(i = 0; i <= dataLen - divLen; i++)
    {
        if(temp[i] == '1')
        {
            for(j = 0; j < divLen; j++)
            {
                temp[i + j] = ((temp[i + j] - '0') ^ (divisor[j] - '0')) +
'0';
            }
        }
    }

    /* Check remainder */

    for(i = dataLen - divLen + 1; i < dataLen; i++)
    {
        if(temp[i] != '0')
        {
            error = 1;
```

```
            break;
        }
    }

    if(error)
        printf("\n\nError Detected.");
    else
        printf("\n\nNo Error Detected.");

    getch();
}
```

Output: